

Haorui Wang

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Education

University of Michigan - Ann Arbor, MI | GPA: 3.7/4.0 University Honors

Aug. 2023 - May 2026

- Bachelor of Science in Data Science

Work Experience

Student Researcher for Trinity Health, Integrated Health Associates, Inc.

Jan. 2025 - Present

University of Michigan | Multidisciplinary Design Program

- Applied NLP, xGBoost, SVM, Random Forest, Logistic Regression, and other ML techniques to analyze medical records and billing data, achieving an accuracy $\geq 70\%$.
- Engineered and fine-tuned IHAGPT (LLM-based) prompt pipelines on Microsoft Fabric, achieving $\geq 80\%$ classification accuracy on audited healthcare records.
- Assisted the business analysis team in detecting billing errors, predicting improper coding, and enhancing accuracy in healthcare auditing, addressing an estimated \$10–\$20 billion annual cost in billing mistakes.

Data Analysis Intern

Jun. 2024 - Aug. 2024

Haier Global Business Co.Ltd

- Analyzed 500,000 Amazon sales records using a Python-generated Evaluation Metric to support market research for small appliances.
- Contributed significantly to a 58% year-over-year growth in sales of the company's sealing machines and juicers to achieve top 3 rankings in their respective categories on Amazon.

Research Experience

Using Machine Learning for Bias Correction in Weather Forecasts

Aug. 2024 - May. 2025

- Trained ConvGNP model with Pytorch and DeepSensor Python package with archived Climate Forecast System (CFS) data from 2012-2024 on GPU-accelerated, high-performance virtual environments.

Implemented and integrated advanced Bias Correction techniques, including Gaussian blob, Time-Varying, Spatially Varying, and Random Noise biases, to rigorously test model robustness, enhancing predictive capabilities for climate-sensitive decision-making.

Professional Experience

Multi-AI-Agent Quantitative Trading Algorithm Development

Mar. 2025 - Present

- Designed and implemented a modular, LLM-driven multi-agent system via API (ChatGPT 4o & Qwen & Yi) to perform natural language-based reasoning, news analysis, and real-time debate coordination based on daily and intraday market signals for quantitative trading in the A-share market.
- Building full-stack ML workflows for data preprocessing, agent communication, and execution automation, integrating prompt engineering, multi-turn reasoning, and backtesting to refine the model.

AI Design Generator Development

Jun. 2025 - Present

- Designed and implemented a Flask-based web application that generates AI-powered image designs using Replicate API (ChatGPT 4o & DALL · E).
- Engineering a dual-prompt pipeline that separates design briefs for editable layer metadata and optimized visual prompts for image generation. Enabled UI friendly editing through AI-assisted layer separation.

Stock Market Prediction and Sentiment Analysis Project

Jul. 2024 - Aug. 2024

- Conducted Sentiment Analysis for the top 25 daily Reddit headlines using VADER and TextBlob, integrated with historical Dow information from 2008 to 2016.
- Trained and evaluated ML models using Scikit-learn packages such as Naive Bayes, SVM, Random Forest, and Logistic Regression, achieving approximately 60% accuracy on market movement prediction.

Skills & Certifications

- **Technical skills:** Python, C++; Machine Learning & AI, Model Training; Data Analysis, FinTech (Blockchain, Cryptocurrency, Stocks Trading, Quantitative Analysis).
- **Certification:** IBM Machine Learning Professional Certificate; Google Data Analytics Professional Certificate;